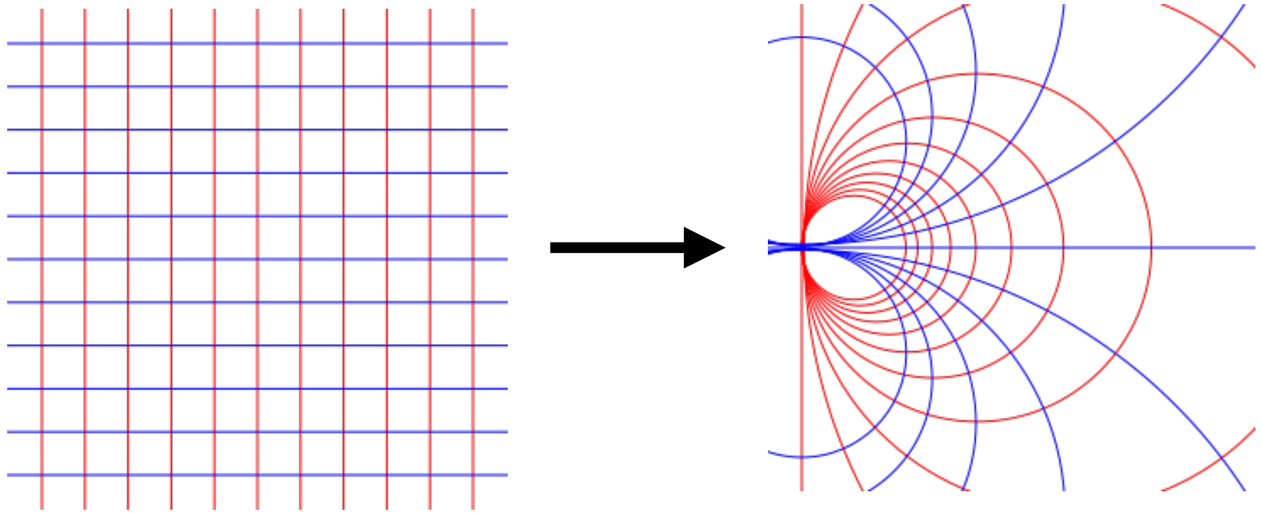
Conformal invariance

- I will now describe the first project of my **Pivot Fellowship**: how to make a neural network that is equivariant under **conformal invariance.**

- Recall conformal invariance is the set of coordinate transformations that **preserve angles**. In d dimensions, it is the group $SO(d+1,1)$.



$$x \to Rx + t$$ **Translations, rotations**

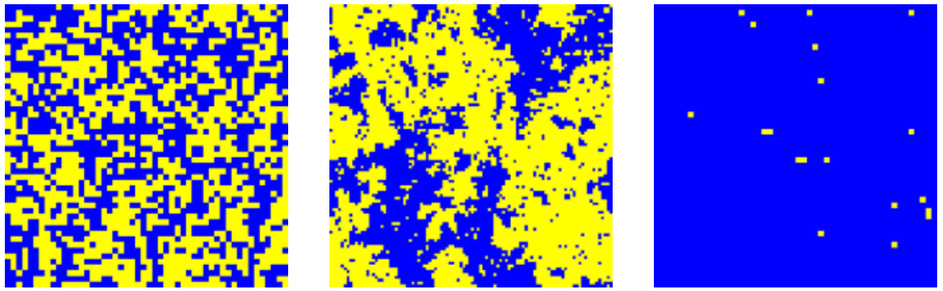
$$x \to \lambda x$$ **Scaling**

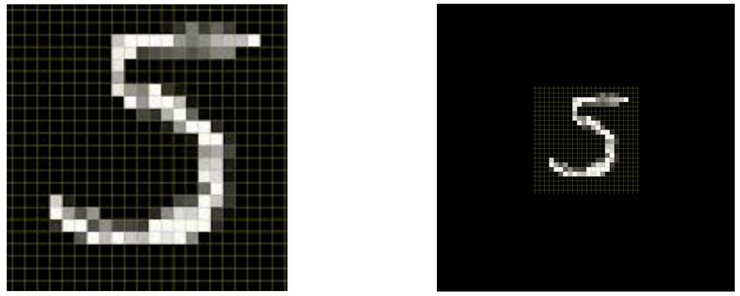
$$\frac{x}{|x|^2} \to \frac{x}{|x|^2} + b$$ **Special conformal**

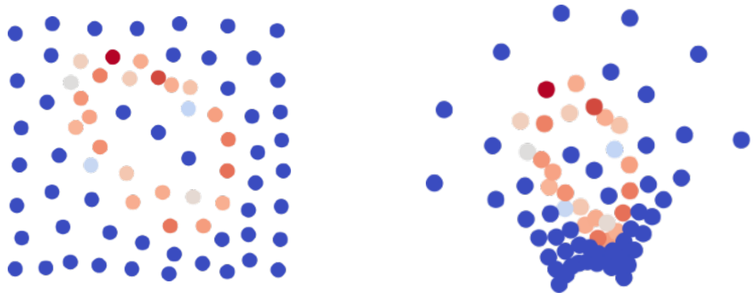
Uses of scale and conformal invariance

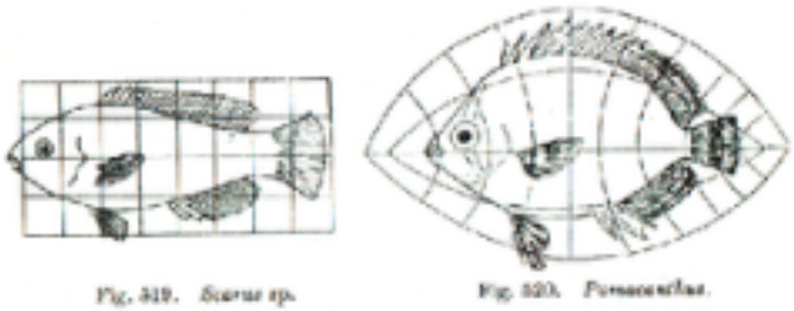
- Conformal invariance plays an important role in physics. Applications range from phase transitions — e.g. the Curie point in magnets — to quantum gravity (this will actually be relevant later!)

- In computer and human vision there is a useful notion of scale (and possibly conformal) invariance (e.g. Lei et. al 2023).









Of Growth and Form, Thompson 1917